

1. What is a Database?

Answer

Database-na related data-ah store, manage, retrieve panna use aagura system.

Company-la employee details, customer details, orders, salary records maari information-ah organized-a store panna database use pannuvanga.

Example

College database-la:

Student Details
Marks
Attendance
Faculty Details

store pannalam.

Interview Line

"Database is a collection of related data used to store and manage information efficiently."

2. What is SQL?

Answer

SQL-na Structured Query Language.

Database-kitta communicate panna use pannura language. SQL use panni data retrieve, insert, update, delete panna mudiyum.

Example

```
SELECT * FROM Employees;
```

Indha query Employees table-la iruka ella records-um kaatum.

Interview Line

"SQL is a language used to interact with relational databases."

3. What is RDBMS?

Answer

RDBMS-na Relational Database Management System.

Idhu data-ah tables form-la store pannum. Oru table-ah innoru table-oda connect panna relationships use pannuvom.

Example

Employee Table

Department Table

Foreign Key use panni rendayum connect pannalam.

Interview Line

"RDBMS stores data in tables and maintains relationships between them."

4. Examples of RDBMS?

Answer

Industry-la adhigama use panna RDBMS:

MySQL
Microsoft SQL Server
Oracle
PostgreSQL

Iva ellam SQL support pannum.

Interview Line

"Popular RDBMS examples are SQL Server, Oracle, MySQL and PostgreSQL."

5. What is a Table?

Answer

Table-na database-la data store panna basic structure.

Data rows and columns form-la organize aagum.

Example

EmpId	Name	Salary
101	Ravi	30000

Idhu oru Employee table.

Interview Line

"A table is a collection of rows and columns used to store data."

6. What is a Row?

Answer

Row-na table-la iruka single record.

Oru employee pathina complete information oru row-la irukkum.

Example

101 | Ravi | 30000

Idhu oru row.

Interview Line

"A row represents a single record in a table."

7. What is a Column?

Answer

Column-na data-oda attribute or property.

Oru column-la same type information store aagum.

Example

Employee table-la:

EmpId
Name
Salary
Department

Iva ellam columns.

Interview Line

"A column represents a specific attribute of data."

8. What is a Primary Key?

Answer

Primary Key-na table-la ovvoru record-ah unique-a identify panna use pannura column.

Duplicate values-um NULL values-um allow pannaadhu.

Example

EmpId

Employee ID unique-a irukkanum.

Benefits

Duplicate records avoid pannum
Data integrity maintain pannum

Interview Line

"Primary Key uniquely identifies each record in a table."

9. Can Primary Key contain NULL?

Answer

No.

Primary Key mandatory-a value vechirukkanum.

NULL value irundha record-ah unique-a identify panna mudiyadhu.

Example

Valid:

101
102
103

Invalid:

101
NULL
103

Interview Line

"Primary Key cannot contain NULL values."

10. What is a Foreign Key?

Answer

Foreign Key-na oru table-ah innoru table-oda connect panna use pannura column.

Idhu another table Primary Key-ah reference pannum.

Example

Department Table

DeptId	DeptName
1	HR
2	IT

Employee Table

EmpId	Name	DeptId
101	Ravi	1

Inga Employee table-la iruka DeptId dhaan Foreign Key.

Benefits

Relationship maintain pannum
Invalid data prevent pannum

Interview Line

"Foreign Key is used to establish relationships between tables."

11. What is SELECT?

Answer

SELECT statement database-la irundhu data retrieve panna use pannuvom.

Most commonly use panna SQL command idhu dhaan.

Example

```
SELECT * FROM Employees;
```

Interview Line

"SELECT is used to retrieve data from a table."

12. What is WHERE?

Answer

WHERE clause specific records mattum filter panna use pannuvom.

Condition satisfy panna records mattum return aagum.

Example

```
SELECT *  
FROM Employees  
WHERE Salary > 30000;
```

Interview Line

"WHERE clause is used to filter records based on a condition."

13. What is ORDER BY?

Answer

ORDER BY data-ah ascending illa descending order-la sort panna use pannuvom.

Reports generate pannumbodhu adhigama use aagum.

Example

```
SELECT *  
FROM Employees  
ORDER BY Salary DESC;
```

Interview Line

"ORDER BY is used to sort data."

14. What is DISTINCT?

Answer

DISTINCT duplicate values remove pannum.

Unique values mattum display pannum.

Example

```
SELECT DISTINCT Department
FROM Employees;
```

Interview Line

"DISTINCT returns only unique values."

15. Difference between DELETE and DROP?

DELETE

Table structure irukkum.

Rows mattum remove aagum.

```
DELETE FROM Employees;
```

DROP

Table structure-um data-um complete-a remove aagum.

```
DROP TABLE Employees;
```

Interview Line

"DELETE removes records, whereas DROP removes the entire table."

16. COUNT() use?

Answer

COUNT() function table-la iruka records-oda count kandupidikka use pannuvom.

Idhu total rows count panna, specific column values count panna, report generation-ku romba useful.

Example

```
SELECT COUNT(*)
FROM Employees;
```

Interview Line

"COUNT() function is used to calculate the number of records in a table."

17. SUM() use?

Answer

SUM() function numeric values-oda total calculate panna use pannuvom.

Salary, sales amount, marks maari values total kandupidikka use

pannuvanga.

Example

```
SELECT SUM(Salary)
FROM Employees;
```

Interview Line

"SUM() function is used to calculate the total of a numeric column."

18. AVG() use?

Answer

AVG() function numeric values-oda average calculate panna use pannuvom.

Company-la average salary, average sales maari reports generate panna use pannuvanga.

Example

```
SELECT AVG(Salary)
FROM Employees;
```

Interview Line

"AVG() function is used to calculate the average value of a numeric column."

19. MAX() use?

Answer

MAX() function oru column-la iruka highest value kandupidikka use pannuvom.

Highest salary, highest mark, highest sales kandupidikka useful.

Example

```
SELECT MAX(Salary)
FROM Employees;
```

Interview Line

"MAX() function returns the highest value from a column."

20. MIN() use?

Answer

MIN() function oru column-la iruka lowest value kandupidikka use pannuvom.

Lowest salary, lowest mark, minimum sales maari values retrieve panna use pannuvom.

Example

```
SELECT MIN(Salary)
FROM Employees;
```

Interview Line

"MIN() function returns the smallest value from a column."

GROUP BY & HAVING

21. What is GROUP BY?

Answer

GROUP BY clause similar values iruka records-ah group panna use pannuvom.

Department-wise reports, category-wise reports generate panna GROUP BY use pannuvanga.

Example

```
SELECT Department,
               COUNT(*)
FROM Employees
GROUP BY Department;
```

Interview Line

"GROUP BY is used to group similar records together for analysis and reporting."

22. What is HAVING?

Answer

HAVING clause grouped data-ah filter panna use pannuvom.

WHERE individual rows-ah filter pannum, HAVING groups-ah filter pannum.

Example

```
SELECT Department,
               COUNT(*)
FROM Employees
GROUP BY Department
HAVING COUNT(*) > 5;
```

Interview Line

"HAVING is used to filter grouped records after GROUP BY."

23. Difference between WHERE and HAVING?

Answer

WHERE clause individual rows-ah filter pannum.

HAVING clause GROUP BY mudincha apram grouped results-ah filter pannum.

Example

```
SELECT Department,
       COUNT(*)
FROM Employees
WHERE Salary > 30000
GROUP BY Department
HAVING COUNT(*) > 2;
```

Interview Line

"WHERE filters rows before grouping, whereas HAVING filters groups after grouping."

24. Can HAVING work without GROUP BY?

Answer

Yes.

Technically HAVING GROUP BY illaamalum use panna mudiyum, aana practical-a mostly GROUP BY-oda dhaan use pannuvom.

Example

```
SELECT COUNT(*)
FROM Employees
HAVING COUNT(*) > 10;
```

Interview Line

"Yes, HAVING can work without GROUP BY, but it is commonly used with GROUP BY."

25. Why GROUP BY use pannuvom?

Answer

Data-ah category-wise summarize panna GROUP BY use pannuvom.

Reports generate panna idhu romba important.

Example

Department-wise:

```
Employee Count
Total Salary
Average Salary
```

Interview Line

"GROUP BY is used to generate summarized reports from large datasets."

26. What is JOIN?

Answer

JOIN multiple tables-la iruka related data-ah combine panna use pannuvom.

Real-time applications-la data multiple tables-la store aagum, so JOIN romba important.

Example

Employee Table + Department Table

Interview Line

"JOIN is used to retrieve related data from multiple tables."

27. What is INNER JOIN?

Answer

INNER JOIN rendu tables-la matching records mattum return pannum.

Matching data illa na result-la varaadhu.

Example

```
SELECT *  
FROM Employees e  
INNER JOIN Departments d  
ON e.DeptId = d.DeptId;
```

Interview Line

"INNER JOIN returns only matching records from both tables."

28. What is LEFT JOIN?

Answer

LEFT JOIN left table-la iruka ellaa records-um return pannum.

Right table-la matching record illa na NULL values return aagum.

Example

```
SELECT *  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerId = o.CustomerId;
```

Interview Line

"LEFT JOIN returns all records from the left table and matching records from the right table."

29. Difference between INNER JOIN and LEFT JOIN?

Answer

INNER JOIN matching rows mattum return pannum.

LEFT JOIN left table records ellam return pannum, match illa na NULL kaatum.

Interview Line

"INNER JOIN returns only matched records, whereas LEFT JOIN returns all records from the left table."

30. Why JOIN use pannuvom?

Answer

Database-la data multiple tables-la store pannuvom.

Andha related data-ah combine panni meaningful reports generate panna JOIN use pannuvom.

Example

Customer Name + Order Details.

Interview Line

"JOIN helps retrieve related information stored across multiple tables."

31. What is a View?

Answer

View-na virtual table.

Complex query-ah save panni reuse panna use pannuvom.

Example

```
CREATE VIEW ITEmployees
AS
SELECT *
FROM Employees
WHERE Department = 'IT';
```

Interview Line

"A View is a virtual table based on the result of a SQL query."

32. Does View store data?

Answer

Normally View data store pannaadhu.

View query definition mattum store pannum.

Data actual table-la dhaan irukkum.

Interview Line

"No, a View stores the query definition, not the actual data."

33. Why View use pannuvom?

Answer

Security improve panna.

Complex queries simplify panna.

Repeated queries avoid panna.

Interview Line

"Views improve security, simplify complex queries and increase reusability."

34. What is Index?

Answer

Index database-la search speed improve panna use pannura object.

Book-la index page maari work pannum.

Example

EmployeeId column-la search frequent-a nadandha index create pannuvom.

Interview Line

"An Index improves data retrieval performance."

35. Advantage of Index?

Answer

Search fast aagum.

Query performance improve aagum.

Large tables-la response time reduce aagum.

Interview Line

"Indexes help retrieve data faster by reducing the number of rows scanned."

36. What is Stored Procedure?

Answer

Stored Procedure-na database-la save panna SQL program.

Multiple SQL statements-ah oru block-a save panni execute panna mudiyum.

Interview Line

"A Stored Procedure is a precompiled SQL program stored in the database."

37. Why Stored Procedure use pannuvom?

Answer

Code reuse panna.

Security improve panna.

Performance improve panna.

Interview Line

"Stored Procedures improve reusability, security and performance."

38. How to execute Procedure?

Answer

EXEC keyword use panni execute pannuvom.

Example

EXEC GetEmployees;

Interview Line

"Stored Procedures are executed using the EXEC command."

39. Can Procedure accept parameters?

Answer

Yes.

Input values receive panni dynamic results generate panna mudiyum.

Example

EXEC GetEmployeeById 101;

Interview Line

"Yes, Stored Procedures can accept input parameters."

40. Procedure advantage?

Answer

Business logic centralize panna mudiyum.

Repeated SQL code avoid panna mudiyum.

Maintenance easy aagum.

Interview Line

"Stored Procedures reduce code duplication and improve maintainability."

41. What is Function?

Answer

Function-na value return pannura database object.

Calculations perform panna frequently use pannuvom.

Interview Line

"A Function is a database object that returns a value."

42. Function return value mandatory ah?

Answer

Yes.

Function-oda main purpose value return pannuradhu.

Adhanala return type mandatory.

Interview Line

"Yes, every function must return a value."

43. Difference between Procedure and Function?

Answer

Procedure tasks execute panna use pannuvom.

Function value return panna use pannuvom.

Procedure EXEC use pannum.

Function SELECT statement-la use panna mudiyum.

Interview Line

"Procedures perform actions, whereas Functions return values."

44. What is Trigger?

Answer

Trigger-na automatic-a execute aagura database object.

Specific event nadandha udane execute aagum.

Example

Employee insert pannumbodhu audit table update pannalam.

Interview Line

"A Trigger executes automatically when a specified database event occurs."

45. Trigger eppo execute aagum?

Answer

INSERT, UPDATE, DELETE operations nadakkumbodhu execute aagum.

Database events-ku automatic response kudukkum.

Interview Line

"Triggers execute automatically during INSERT, UPDATE or DELETE operations."

46. What is T-SQL?

Answer

T-SQL (Transact-SQL) is Microsoft SQL Server-oda extended version of SQL.

Variables, loops, conditions, error handling maari programming features support pannum.

Interview Line

"T-SQL extends SQL by adding programming capabilities such as variables and control statements."

47. What is DECLARE?

Answer

DECLARE keyword variable create panna use pannuvom.

Temporary values store panna useful.

Example

```
DECLARE @Salary INT;
```

Interview Line

"DECLARE is used to create variables in T-SQL."

48. What is IF ELSE?

Answer

Condition-based decisions edukka use pannuvom.

Condition true-na IF block execute aagum, false-na ELSE block execute aagum.

Interview Line

"IF ELSE is used to implement conditional logic in T-SQL."

49. What is Normalization?

Answer

Normalization-na database design process.

Duplicate data reduce panni data consistency improve panna use pannuvom.

Benefits

- Less redundancy
- Better integrity
- Easier maintenance

Interview Line

"Normalization is the process of organizing data to reduce redundancy and improve consistency."

50. What are 1NF, 2NF, 3NF?

Answer

1NF (First Normal Form)

Repeating groups remove pannuvom.

Oru cell-la oru value mattum irukkanum.

2NF (Second Normal Form)

Partial dependency remove pannuvom.

Non-key columns full primary key mela depend aaganum.

3NF (Third Normal Form)

Transitive dependency remove pannuvom.

Non-key columns innoru non-key column mela depend aaga koodadhu.

Interview Line

"1NF removes repeating groups, 2NF removes partial dependency and 3NF removes transitive dependency."

Intermediate SQL Interview Questions (51 to 75)

51. Difference between CHAR and VARCHAR?

Answer

CHAR and VARCHAR rendu perum character data store panna use aagura datatypes.

CHAR fixed length datatype. VARCHAR variable length datatype.

Example

CHAR(10)

VARCHAR(10)

CHAR(10)-la "Ravi" store panninaalum 10 characters space occupy pannum.

VARCHAR(10)-la "Ravi" store pannina 4 characters-ku mattum space use aagum.

Interview Line

"CHAR uses fixed storage, whereas VARCHAR uses storage based on actual data length."

52. Difference between DELETE, TRUNCATE and DROP?

Answer

DELETE table-la iruka records-ah remove pannum.

TRUNCATE table-la iruka ellaa records-ah remove pannum but structure retain pannum.

DROP table structure-um data-um complete-a remove pannum.

Example

DELETE FROM Employees;

TRUNCATE TABLE Employees;

DROP TABLE Employees;

Interview Line

"DELETE removes rows, TRUNCATE removes all rows faster, and DROP removes the entire table."

53. What happens if Primary Key is duplicated?

Answer

Primary Key duplicate values allow pannaadhu.

Duplicate value insert panna try pannina database error throw pannum.

Idhu uniqueness maintain panna help pannudhu.

Example

101
101

Invalid because Primary Key must be unique.

Interview Line

"Duplicate Primary Key values are not allowed because every record must be uniquely identified."

54. Can a table have multiple Primary Keys?

Answer

Oru table-ku only one Primary Key dhaan define panna mudiyum.

Aana andha Primary Key multiple columns combine panni create panna mudiyum. Adha Composite Primary Key nu solluvom.

Interview Line

"A table can have only one Primary Key, but that key can consist of multiple columns."

55. What is Composite Key?

Answer

Two or more columns serndhu oru record-ah uniquely identify pannina adhu Composite Key.

Single column uniqueness kudukka mudiyadha situation-la use pannuvom.

Example

StudentID + CourseID

Interview Line

"Composite Key is a combination of multiple columns used to uniquely

identify records."

56. What is Candidate Key?

Answer

Candidate Key-na Primary Key aaga eligible-a iruka column illa column set.

Candidate Keys-la irundhu oru key Primary Key-a select pannuvom.

Example

Employee table-la:

EmployeeID
Email

Rendume unique-na rendu perum Candidate Keys.

Interview Line

"Candidate Keys are columns that can uniquely identify records and are eligible to become Primary Keys."

57. What is Unique Key?

Answer

Unique Key duplicate values allow pannaadhu.

Table-la uniqueness maintain panna use pannuvom.

Primary Key maari work pannum but generally NULL values allow pannalaam.

Interview Line

"Unique Key ensures uniqueness of data but is different from Primary Key in handling NULL values."

58. Difference between Primary Key and Unique Key?

Answer

Primary Key table-oda main identifier.

Primary Key NULL allow pannaadhu.

Unique Key duplicate values allow pannaadhu but generally NULL values allow pannalaam.

Interview Line

"Primary Key uniquely identifies records and cannot contain NULL, whereas Unique Key also ensures uniqueness but may allow NULL values."

59. Why Foreign Key use pannuvom?

Answer

Foreign Key two tables-ku relationship create panna use pannuvom.

Idhu invalid data insert aaguradha prevent pannum.

Database consistency maintain panna help pannum.

Example

Employee table DeptId

Department table DeptId

Interview Line

"Foreign Key is used to maintain relationships and ensure data integrity between tables."

60. What is Referential Integrity?

Answer

Related tables-la valid relationship maintain pannuradhu Referential Integrity.

Foreign Key values corresponding parent table-la irukkanum.

Invalid references allow pannaadhu.

Example

Department table-la DeptId 10 illa na Employee table-la DeptId 10 insert panna mudiyadhu.

Interview Line

"Referential Integrity ensures that relationships between tables remain valid and consistent."

61. Why SELECT * avoid pannanum?

Answer

SELECT * use pannina unnecessary columns-um retrieve aagum.

Network traffic increase aagum.

Large tables-la performance reduce aagum.

Required columns mattum retrieve pannina query efficient-a work pannum.

Interview Line

"SELECT * should be avoided because it retrieves unnecessary data and impacts performance."

62. Which executes first? WHERE, GROUP BY, HAVING, ORDER BY

Answer

SQL execution order:

WHERE
GROUP BY
HAVING
ORDER BY

Database first records filter pannum.

Apram grouping pannum.

Apram groups filter pannum.

Finally sort pannum.

Interview Line

"WHERE executes before grouping, HAVING executes after grouping, and ORDER BY executes at the end."

63. Difference between COUNT(*) and COUNT(Column)?

Answer

COUNT(*) table-la iruka ellaa rows-um count pannum.

COUNT(Column) specific column-la NULL illaadha values mattum count pannum.

Example

COUNT(*)
COUNT(Salary)

Interview Line

"COUNT(*) counts all rows, whereas COUNT(Column) ignores NULL values."

64. Difference between = and LIKE?

Answer

= exact value compare panna use pannuvom.

LIKE pattern matching-ku use pannuvom.

Search operations-la LIKE adhigama use aagum.

Example

```
WHERE Name = 'Ravi'  
WHERE Name LIKE 'R%'
```

Interview Line

"= is used for exact matching, whereas LIKE is used for pattern-based searching."

65. Meaning of LIKE 'R%'?

Answer

R letter-la start aagura values ellam retrieve aagum.

Example

```
Ravi  
Raj  
Ramesh
```

Interview Line

""R%" returns values that start with the letter R."

66. Meaning of LIKE '%R'?

Answer

R letter-la end aagura values retrieve aagum.

Example

```
KumarR  
SivaR
```

Interview Line

""%R" returns values that end with the letter R."

67. Meaning of LIKE '%R%'?

Answer

Value-oda yendha position-la irundhaalum R irundha result-la varum.

Example

```
Ravi  
Kumar  
Arjun
```

Interview Line

""%R%" returns values that contain the letter R anywhere in the string."

68. Can INNER JOIN return unmatched rows?

Answer

No.

INNER JOIN matching records mattum return pannum.

Match illaadha rows result-la varaadhu.

Interview Line

"INNER JOIN returns only records that have matching values in both tables."

69. Why LEFT JOIN use pannuvom?

Answer

Left table-la iruka ellaa records-um kaatanum-na LEFT JOIN use pannuvom.

Right table-la match illa na NULL values return aagum.

Interview Line

"LEFT JOIN is used when all records from the left table must be displayed regardless of matching records."

70. Which join use panni customers without orders kandupudikalam?

Answer

LEFT JOIN use pannuvom.

LEFT JOIN pannitu Order table column NULL iruka records filter pannina customers without orders kedaikkum.

Example

```
SELECT c.CustomerName
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

Interview Line

"LEFT JOIN with NULL checking helps identify customers who have not placed orders."

71. Can data be updated through a View?

Answer

Simple Views-la update panna mudiyum.

Aana JOIN, GROUP BY, Aggregate Functions use panna complex Views-la update panna restrictions irukkum.

Interview Line

"Simple views are updatable, whereas complex views are generally not."

72. Why Stored Procedures faster?

Answer

Stored Procedures execution plan cache aagum.

Same procedure repeated-a execute pannumbodhu database marubadiyum compile panna thevai illa.

Idhu performance improve pannum.

Interview Line

"Stored Procedures are faster because their execution plans can be reused."

73. Can Procedure call another Procedure?

Answer

Yes.

Oru Stored Procedure-ku ulla innoru Stored Procedure execute panna mudiyum.

Large applications-la modular coding-ku idhu useful.

Interview Line

"Stored Procedures can call other procedures to improve code reusability and maintainability."

74. Can Function modify table data?

Answer

Generally SQL Functions data return panna use pannuvom.

Functions-la INSERT, UPDATE, DELETE operations panna restrictions irukkum.

Interview Line

"Functions are primarily designed to return values and generally should not modify table data."

75. Trigger disadvantage?

Answer

Trigger automatic-a background-la execute aagum.

Adhanala debugging kashtam.

Excessive triggers use pannina database performance impact aagalam.

Interview Line

"Triggers can make debugging difficult and may affect performance if overused."

76. Employee table-la 1 lakh records irukku. Search slow-a irukku. Enna pannuva?

Answer

Employee table-la lakhs of records irukkumbodhu search operation slow aaga chance irukku.

Mudhala search adhigama nadakkura columns (EmpId, Email, DepartmentId) mela Index create pannuven.

Index use pannina database full table scan pannaama direct-a required records locate pannum.

Query performance improve panna unnecessary columns avoid pannuvom, proper WHERE condition use pannuvom.

Example

```
CREATE INDEX idx_empid  
ON Employees(EmpId);
```

Interview Line

"If search performance is slow, I would analyze the query and create appropriate indexes on frequently searched columns."

77. Department-wise salary report generate panna enna use pannuva?

Answer

Department-wise report generate panna GROUP BY use pannuvom.

GROUP BY similar department records-ah group pannum. Adhoda SUM() function use panni total salary calculate panna mudiyum.

Idhu reporting and management dashboards-la romba common-a use aagura query.

Example

```

SELECT Department,
       SUM(Salary) AS TotalSalary
FROM Employees
GROUP BY Department;

```

Output

```

Department  TotalSalary
HR          55000
IT          80000

```

Interview Line

"Department-wise reports are generated using GROUP BY along with aggregate functions such as SUM, AVG or COUNT."

78. Duplicate employee records kandupudikka?

Answer

Duplicate records identify panna GROUP BY and HAVING use pannuvom.

GROUP BY similar values-ah group pannum. HAVING COUNT(*) > 1 condition use panni duplicate records kandupidikkalam.

Data cleaning and validation process-la idhu frequently use pannuvanga.

Example

```

SELECT Name,
       COUNT(*)
FROM Employees
GROUP BY Name
HAVING COUNT(*) > 1;

```

Interview Line

"Duplicates can be identified using GROUP BY and HAVING COUNT(*) greater than 1."

79. Second highest salary kandupudikka enna approach?

Answer

Second highest salary kandupidikka multiple approaches irukku.

Most common fresher-level approach Subquery use pannuradhu.

Mudhala highest salary retrieve pannuvom. Apram adha exclude panni remaining salaries-la MAX() apply pannuvom.

Example

```

SELECT MAX(Salary)
FROM Employees
WHERE Salary <
(
    SELECT MAX(Salary)

```

```
        FROM Employees
    );
```

Alternative Methods

```
TOP with ORDER BY
DENSE_RANK()
ROW_NUMBER()
```

Interview Line

"Second highest salary can be found using a subquery by excluding the highest salary and then finding the maximum from the remaining records."

80. Stored Procedure use panna 3 advantages sollu.

Answer

Stored Procedures enterprise applications-la adhigama use pannuvanga because they provide multiple benefits.

1. Reusability

Oru procedure once create pannina multiple times execute panna mudiyum.

Same SQL code marubadiyum ezhudha thevai illa.

2. Security

Users-ku direct table access kudukkaama procedure execute permission mattum kudukkalaam.

Idhu database security improve pannum.

3. Better Performance

Stored Procedure execution plan cache aagum.

Repeated executions-ku query marubadiyum compile panna thevai illa.

Example

```
CREATE PROCEDURE GetEmployees
AS
BEGIN
    SELECT *
    FROM Employees;
END;
```

Interview Line

"Stored Procedures improve reusability, enhance security and provide better performance through execution plan caching."